

SKILL 22

```
root@PC: ~/os_lab/combined_complex_lab
root@PC:~# cd ~/os_lab/combined_complex_lab
root@PC:~/os_lab/combined_complex_lab# nano modular_shell_error_handling.c
root@PC:~/os_lab/combined_complex_lab# #include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <signal.h>
#include <errno.h>
#include <termios.h>

#define MAX_LINE 1024
#define MAX_ARGS 64
#define MAX_JOBS 32

typedef enum {
    JOB_RUNNING,
    JOB_STOPPED,
    JOB_DONE
} JobState;

typedef struct {
    int job_id;
    pid_t pid;
    pid_t pgid;
    int is_foreground;
    char command[MAX_LINE];
    JobState state;
} Job;

// Global Shell State
Job job_table[MAX_JOBS];
int next_job_id = 1;

pid_t shell_pgid;
struct termios shell_tmodes;
int shell_terminal;

// Safe Async Diagnostic Logger
```

```
root@PC: ~/os_lab/combined_complex_lab
-bash: syntax error near unexpected token `stderr,'
-bash: syntax error near unexpected token `}'
[1] 839
-bash: //: Is a directory
SIGNALS: command not found
[1]+ Exit 126 // --- MODULE 1: ERROR SCENARIO DIAGNOSTICS
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `sig'
Command 'const' not found, did you mean:
  command 'coinst' from deb coinst (1.9.3-5)
Try: apt install <deb name>
-bash: syntax error near unexpected token `STDERR_FILENO,'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
struct: command not found
-bash: syntax error near unexpected token `&'
sa_tstp.sa_handler: command not found
-bash: syntax error near unexpected token `&'
sa_tstp.sa_flags: command not found
-bash: syntax error near unexpected token `SIGTSTP,'
-bash: syntax error near unexpected token `"SignalSetup",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `SIGINT,'
-bash: syntax error near unexpected token `SIGTTIN,'
-bash: syntax error near unexpected token `SIGTTOU,'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
job_table[i].job_id: command not found
job_table[i].pid: command not found
job_table[i].pgid: command not found
job_table[i].is_foreground: command not found
job_table[i].state: command not found
job_table[i].command[0]: command not found
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
```

```
root@PC: ~/os_lab/combined_complex_lab
void log_error(const char *module, const char *msg) {
    fprintf(stderr, "[ERROR - %s]: %s\n", module, msg);
} return 0;te_pipeline(args, raw_cmd, is_background);d>");ound);");\n", jid, pid);
typedef: command not found
JOB_RUNNING,: command not found
JOB_STOPPED,: command not found
JOB_DONE: command not found
-bash: syntax error near unexpected token `}'
typedef: command not found
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
pid_t: command not found
Command 'int' not found, but there are 29 similar ones.
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'pchar' from deb pchar (1.5-6)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'shar' from deb sharutils (1:4.15.2-9)
See 'snap info <snapname>' for additional versions.
JobState: command not found
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
Command 'Job' not found, did you mean:
  command 'mob' from snap mob-sh (4.2.0)
  command 'wob' from deb wob (0.14.2-1)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
struct: command not found
Command 'int' not found, but there are 29 similar ones.
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `stderr,'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `stderr,'
```

```
root@PC: ~/os_lab/combined_complex_lab
job_table[i].state: command not found
job_table[i].command[0]: command not found
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{
job_table[i].job_id: command not found
job_table[i].pid: command not found
job_table[i].pgid: command not found
job_table[i].is_foreground: command not found
-bash: syntax error near unexpected token `job_table[i].command,'
job_table[i].state: command not found
-bash: return: job_table[i].job_id: numeric argument required
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"JobTable",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
job_table[index].job_id: command not found
job_table[index].pid: command not found
job_table[index].pgid: command not found
job_table[index].is_foreground: command not found
job_table[index].state: command not found
job_table[index].command[0]: command not found
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `"\n%-6s %-8s %-8s %-12s %s\n",'
-bash: syntax error near unexpected token `"-----\n"'
-----\n"
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `?'
-bash: syntax error near unexpected token `"[d]    %-8d %-8d %-12s %s\n",'
job_table[i].job_id,: command not found
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
```

```
root@PC: ~/os_lab/combined_complex_lab
*is_background: command not found
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `line'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `"Parser",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
args[arg_count++]: command not found
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `}'
args[arg_count]: command not found
-bash: syntax error near unexpected token `return'
[1] 1439
-bash: //: Is a directory
operator: command not found
[1]+  Exit 126          // Syntax validation: standalone
-bash: syntax error near unexpected token `args[0],'
-bash: syntax error near unexpected token `"SyntaxError",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `args[arg_count - 1],'
*is_background: command not found
args[arg_count - 1]: command not found
arg_count--: command not found
-bash: syntax error near unexpected token `}'
-bash: return: arg_count: numeric argument required
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
[1] 1509
-bash: //: Is a directory
RECOVERY: command not found
[1]+  Exit 126          // --- MODULE 4: TERMINAL CONTROL
-bash: syntax error near unexpected token `('
shell_terminal: command not found
-bash: syntax error near unexpected token `shell_terminal'
-bash: syntax error near unexpected token `-shell_pgid,'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `<'
```

root@PC: ~/os_lab/combined_complex_lab

```
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"\n"'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `('
Command 'int' not found, but there are 29 similar ones.
pid_t: command not found
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `status'
-bash: syntax error near unexpected token `"[d]+ Terminated with exit status %d\t%s\n",'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `i'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"[d]+ Killed by signal %d\t\t%s\n",'
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `i'
-bash: syntax error near unexpected token `}'
job_table[i].state: command not found
job_table[i].is_foreground: command not found
-bash: syntax error near unexpected token `"[d]+ Suspended (SIGTSTP)\t\t%s\n",'
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
job_table[i].state: command not found
-bash: syntax error near unexpected token `"[d]+ Continued\t\t\t%s\n",'
-bash: syntax error near unexpected token `)'
-bash: syntax error near unexpected token `}'
-bash: break: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
[1] 1348
-bash: //: Is a directory
INTERFACE: command not found
[1]+ Exit 126 // --- MODULE 3: INPUT PARSER
-bash: syntax error near unexpected token `('
*is background: command not found
```

root@PC: ~/os_lab/combined_complex_lab

```
[1] 1580
Command 'Job' not found, did you mean:
  command 'mob' from snap mob-sh (4.2.0)
  command 'wob' from deb wob (0.14.2-1)
See 'snap info <snapname>' for additional versions.
job_table[index]: command not found
[1]+  Exit 127                  Job *job =
job-: command not found
-bash: syntax error near unexpected token `shell_terminal,'
-bash: syntax error near unexpected token `"JobControl",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `{ '
job-: command not found
-bash: syntax error near unexpected token `-'
-bash: syntax error near unexpected token `"JobControl",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `-'
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `"JobControl",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `status'
-bash: syntax error near unexpected token `index'
-bash: syntax error near unexpected token `}'
job-: command not found
job-: command not found
-bash: syntax error near unexpected token `"\n[%d]+ Suspended\t\t%s\n",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `shell_terminal,'
-bash: syntax error near unexpected token `shell_terminal,'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `( '
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `( '
-bash: syntax error near unexpected token `{ '
Command 'index' not found, did you mean:
```

```
root@PC: ~/os_lab/combined_complex_lab
command 'cindex' from deb codesearch (0.0~hg20120502-3ubuntu0.24.04.3)
command 'xindex' from deb texlive-extra-utils (2023.20231207-3)
command 'sindex' from deb biosquid (1.9g+cvs20050121-12)
Try: apt install <deb name>
-bash: break: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `"JobControl",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
[1] 1742
Command 'Job' not found, did you mean:
  command 'mob' from snap mob-sh (4.2.0)
  command 'wob' from deb wob (0.14.2-1)
See 'snap info <snapname>' for additional versions.
job_table[index]: command not found
[1]+  Exit 127                  Job *job =
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `"JobControl",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
job-: command not found
-bash: syntax error near unexpected token `-'
-bash: syntax error near unexpected token `"JobControl",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"[%d]+ Resumed in background: %s &\n",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `('
-bash: syntax error near unexpected token `{
-bash: syntax error near unexpected token `"Execution",'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `{
-bash: //: Is a directory
-bash: syntax error near unexpected token `SIGTSTP,'
-bash: syntax error near unexpected token `SIGINT'
```

```
root@PC: ~/os_lab/combined_complex_lab
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `}'
-bash: //: Is a directory
-bash: syntax error near unexpected token `('
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'shar' from deb sharutils (1:4.15.2-9)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'pchar' from deb pchar (1.5-6)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
See 'snap info <snapname>' for additional versions.
Command 'char' not found, did you mean:
  command 'chr' from snap chr (0.80.4)
  command 'pchar' from deb pchar (1.5-6)
  command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
  command 'phar' from deb php-cli (2:8.2+93ubuntu1)
  command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
  command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
  command 'car' from deb ucommon-utils (7.0.1-0.1)
  command 'shar' from deb sharutils (1:4.15.2-9)
See 'snap info <snapname>' for additional versions.
Command 'int' not found, but there are 29 similar ones.
-bash: syntax error near unexpected token `;'
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `;'
-bash: syntax error near unexpected token `"error_shell> '"
-bash: syntax error near unexpected token `stdout'
-bash: syntax error near unexpected token `line,'
-bash: syntax error near unexpected token `stdin'
-bash: syntax error near unexpected token `"\nSession EOF. Exiting shell gracefully.
\n"'
-bash: break: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `"MainLoop",'
-bash: syntax error near unexpected token `stdin'
-bash: continue: only meaningful in a `for', `while', or `until' loop
```

```
root@PC: ~/os_lab/combined_complex_lab
error_shell>
error_shell: command not found
[ERROR: command not found
-bash: syntax error near unexpected token `newline'
root@PC:~/os_lab/combined_complex_lab# error_shell> fg 99
[ERROR - JobControl]: Job ID specified does not exist.
error_shell>
error_shell: command not found
[ERROR: command not found
-bash: syntax error near unexpected token `newline'
root@PC:~/os_lab/combined_complex_lab# error_shell> sleep 100 &
[1] Background process initialized (PID: 12045)
error_shell> jobs

[JID]  PID      PGID      State      Command
-----
[1]    12045    12045    Running    sleep 100

error_shell> fg 1
sleep 100
^Z
[1]+  Suspended                  sleep 100
error_shell> bg 1
[1]+  Resumed in background: sleep 100 &
[1] 1984
-bash: syntax error near unexpected token `('
error_shell: command not found
error_shell: command not found
[JID]: command not found
[1]+  Exit 127                  error_shell 100 > sleep
-----: command not found
[1]: command not found
error_shell: command not found
-bash: :s^Z: substitution failed
[1]+: command not found
error_shell: command not found
[1] 2129
root@PC:~/os_lab/combined_complex_lab# [1]+: command not found
./test_runner.sh
-bash: ./test_runner.sh: Permission denied
```

```
root@PC: ~/os_lab/combined_complex_lab
command 'phar' from deb php8.3-cli (8.3.6-0ubuntu0.24.04.9)
command 'phar' from deb php-cli (2:8.2+93ubuntu1)
command 'phar' from deb php8.2-cli (8.2.12-1ubuntu2)
command 'shar' from deb sharutils (1:4.15.2-9)
command 'chat' from deb ppp (2.4.9-1+1.1ubuntu1)
command 'car' from deb ucommon-utils (7.0.1-0.1)
command 'pchar' from deb pchar (1.5-6)
See 'snap info <snapname>' for additional versions.
-bash: syntax error near unexpected token `raw_cmd,'
raw_cmd[stcrspn(raw_cmd, \n)]: command not found
-bash: syntax error near unexpected token `('
-bash: //: Is a directory
-bash: syntax error near unexpected token `continue'
-bash: syntax error near unexpected token `args[0],'
-bash: break: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `;'
-bash: continue: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `"SyntaxError",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `atoi'
-bash: syntax error near unexpected token `}'
-bash: continue: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `{ '
-bash: syntax error near unexpected token `"SyntaxError",'
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `atoi'
-bash: syntax error near unexpected token `}'
-bash: continue: only meaningful in a `for', `while', or `until' loop
-bash: syntax error near unexpected token `}'
-bash: syntax error near unexpected token `args,'
-bash: syntax error near unexpected token `}'
-bash: return: can only `return' from a function or sourced script
-bash: syntax error near unexpected token `}'
root@PC:~/os_lab/combined_complex_lab# ./modular_shell_error_handling
-bash: ./modular_shell_error_handling: No such file or directory
root@PC:~/os_lab/combined_complex_lab# ./modular_shell_error_handling
```

```
root@PC: ~/os_lab/combined_complex_lab

error_shell> fg 1
sleep 100
^Z
[1]+  Suspended                  sleep 100
error_shell> bg 1
[1]+  Resumed in background: sleep 100 &
[1] 1984
-bash: syntax error near unexpected token `('
error_shell: command not found
error_shell: command not found
[JID]: command not found
[1]+  Exit 127                  error_shell 100 > sleep
-----: command not found
[1]: command not found
error_shell: command not found
-bash: :s^Z: substitution failed
[1]+: command not found
error_shell: command not found
[1] 2129
root@PC:~/os_lab/combined_complex_lab# [1]+: command not found
./test_runner.sh
-bash: ./test_runner.sh: Permission denied
[1]+  Exit 127                  [1]+  Resumed in background: sleep 100
root@PC:~/os_lab/combined_complex_lab#
```